

1. Abstract

This paper presents a real-time, resource-efficient Speech Emotion Recognition (SER) system trained on the **Toronto Emotional Speech Set (TESS)**, modified to include both male and female voice samples for each emotional label—Angry, Disgust, Fear, Happy, and Neutral—while also incorporating a custom “Noise” class to enhance robustness. The final model was deployed on the M5Stack C Plus, an ESP32-based microcontroller, using TinyML techniques. Unlike conventional cloud-based SER systems that introduce latency and raise privacy concerns, this solution performs entirely offline inference on embedded hardware. Edge Impulse was used to design, train, and optimize a quantized 2D **Convolutional Neural Network (CNN)**, leveraging **Mel-filterbank energy (MFE)** features extracted from 16 kHz audio captured via an SPM1423 PDM digital microphone interfaced over I²S. The model classifies six emotional states with a **validation accuracy of 99.5%** and a total inference latency of **571~572 ms**, including both DSP and classification phases. The final quantized int8 model occupies less than **2 MB** of flash and consumes under **50 KB** of RAM. This work demonstrates that accurate and low-latency emotion recognition is feasible on ultra-low-power microcontrollers, making it ideal for privacy-preserving, always-on, and embedded human-computer interaction systems.

Keywords: Speech Emotion Recognition (SER), Machine Learning, Real-Time Inference, Quantized CNN, Robotics,, Embedded Systems.

2. Introduction

Speech emotion recognition aims to identify affective states from vocal signals. Traditional methods rely on cloud computing, which introduces latency and privacy concerns. In contrast, this work explores deploying SER models on microcontrollers using TinyML, enabling local, secure, and fast inference.

Let $x(t) \in \mathbb{R}$ denote the discrete-time audio signal. The objective is to learn a function

$$f_{\theta} : x(t) \rightarrow y \in \{ y_1, \dots, y_6 \}$$

Where y_i represents one of the six emotion classes, and θ are the learnable parameters of a CNN.

3. Related Work

Prior work in speech emotion recognition has predominantly utilized spectrograms and Mel-frequency cepstral coefficients (MFCCs) as input features, coupled with deep learning architectures such as convolutional or recurrent neural networks. However, most of these approaches are designed for high-resource environments, with limited focus on deployment to resource-constrained edge devices. This study builds upon recent advancements in embedded artificial intelligence—particularly platforms like Edge Impulse and TensorFlow

Lite for Microcontrollers—to enable real-time, on-device emotion recognition in a low-power, memory-limited setting.

4. Methodology

4.1 Data Preprocessing

An 80-20 split was done to divide the dataset into training and test sets. Additionally, a Noise class was created using non-verbal environmental sounds to improve model robustness in real-world conditions.

Class	Total Sample	Training Sample	Test Sample	Total Duration of Time for Training Dataset
Angry	400	322	77	9m 51s
Happy	400	321	79	10m 33s
Disgust	400	320	80	13m 4s
Fear	400	320	79	8m 50s
Neutral	400	320	80	10m 57s
Noise	720	578	142	9m 47s

4.2 Audio Signal Overview

To illustrate the variability of speech patterns across emotional states, representative waveforms for each of the six target classes was plotted. These time-domain plots highlight the amplitude envelope of 1-second clips sampled at 16 kHz.

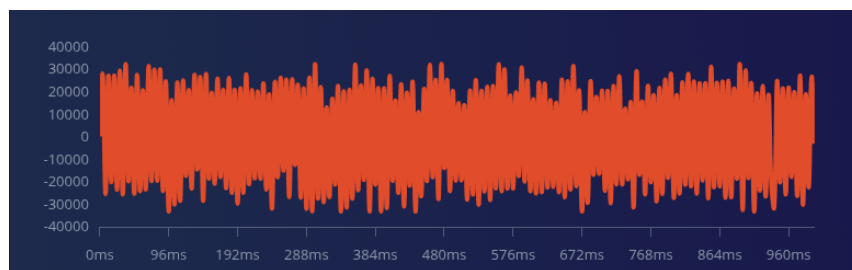


Figure 5: Noise

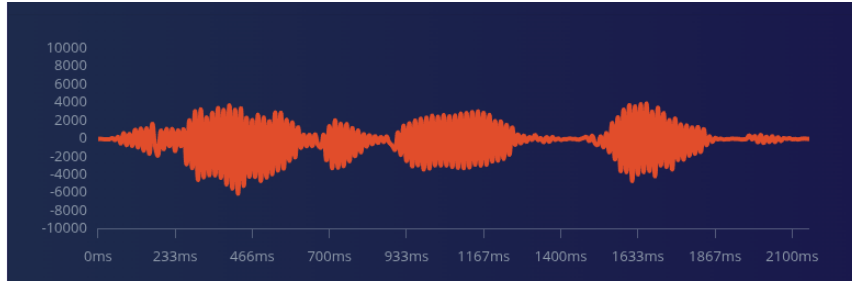


Figure 6: Neutral

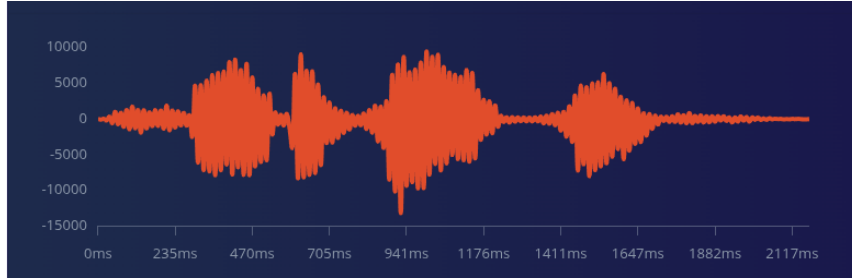


Figure 7: Angry

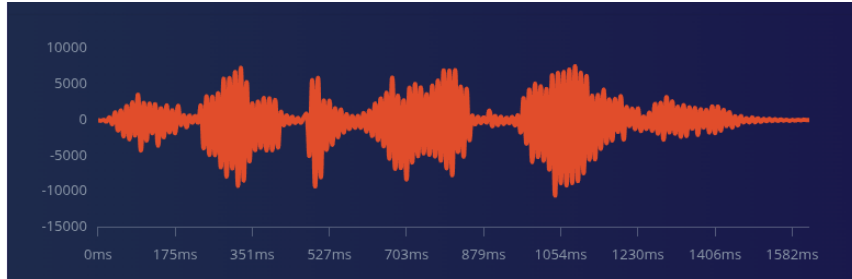


Figure 8: Disgust

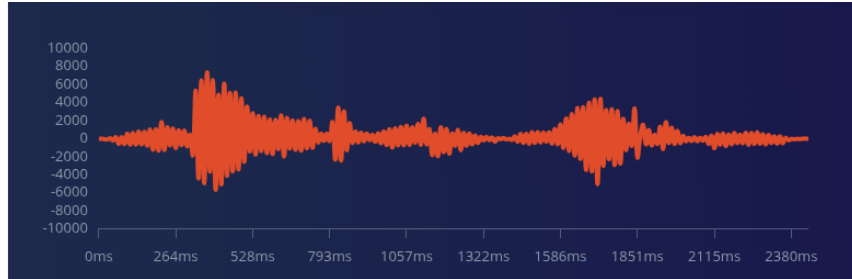


Figure 9: Fear

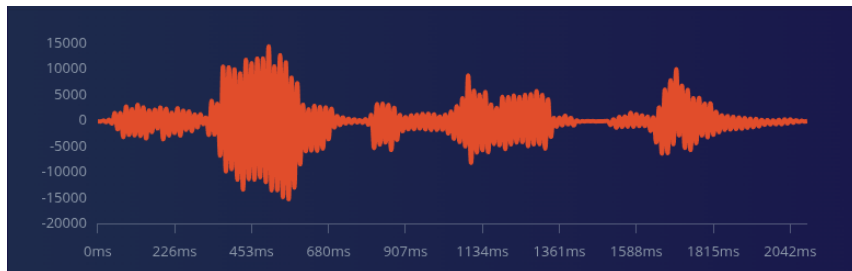


Figure 10: Happy

4.3 Signal Processing and Feature Extraction

The input audio $x(t)$ is segmented into frames of length $T = 32ms$ with a stride of $S = 32ms$. Each frame is transformed using the Short-Time Fourier Transform (STFT) followed by a Mel-filterbank operation:

$$X(k, n) = \sum_{m=0}^{N-1} x(nS+m)w(m)e^{-\frac{j2\pi km}{N}}$$

$$MFE(n, b) = \log \left(\sum_{k=k_b^{min}}^{k_b^{max}} |X(k, n)|^2 \cdot H_b(k) + \epsilon \right)$$

where:

- $w(m)$ is the Hamming window,
- $H_b(k)$ is the b -th triangular Mel filter,
- $\epsilon = 10^{-6}$ is added for numerical stability,
- n is the frame index,
- $b \in \{1, \dots, 32\}$ indexes the 32 filterbanks.

Each 1-second clip yields a **32 X 31** time-frequency matrix of Mel energy features.

Figure 1 below shows the weighting profile applied to FFT bins by the 32 Mel filters:

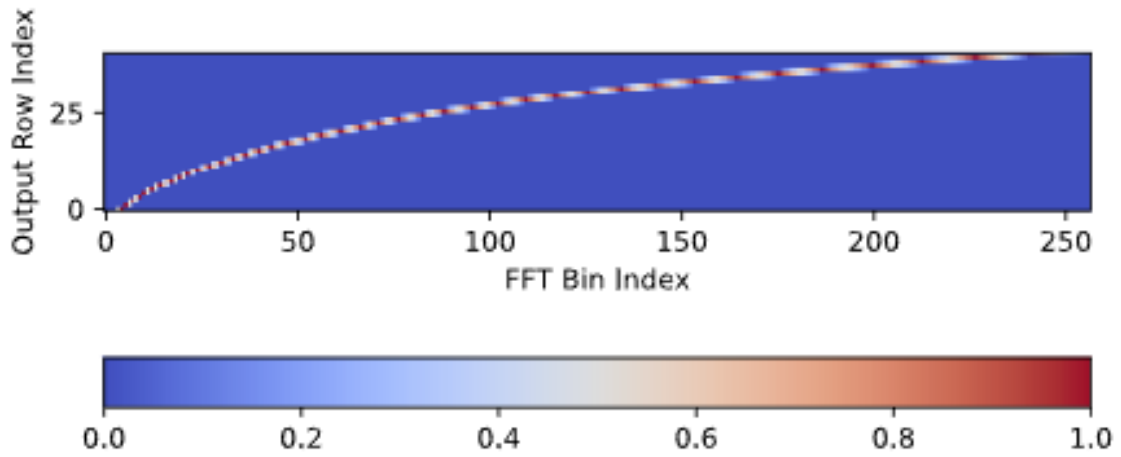


Figure 1

Figure 2 illustrates the resulting Mel energy spectrogram generated from a single 1-second audio sample:

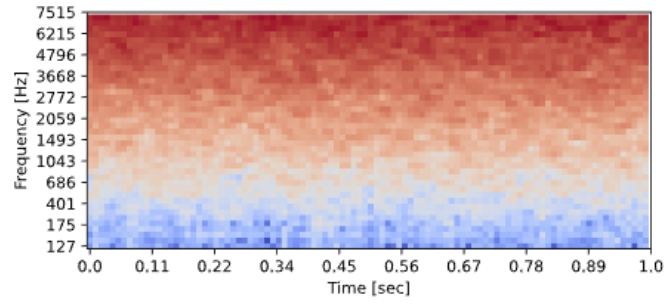


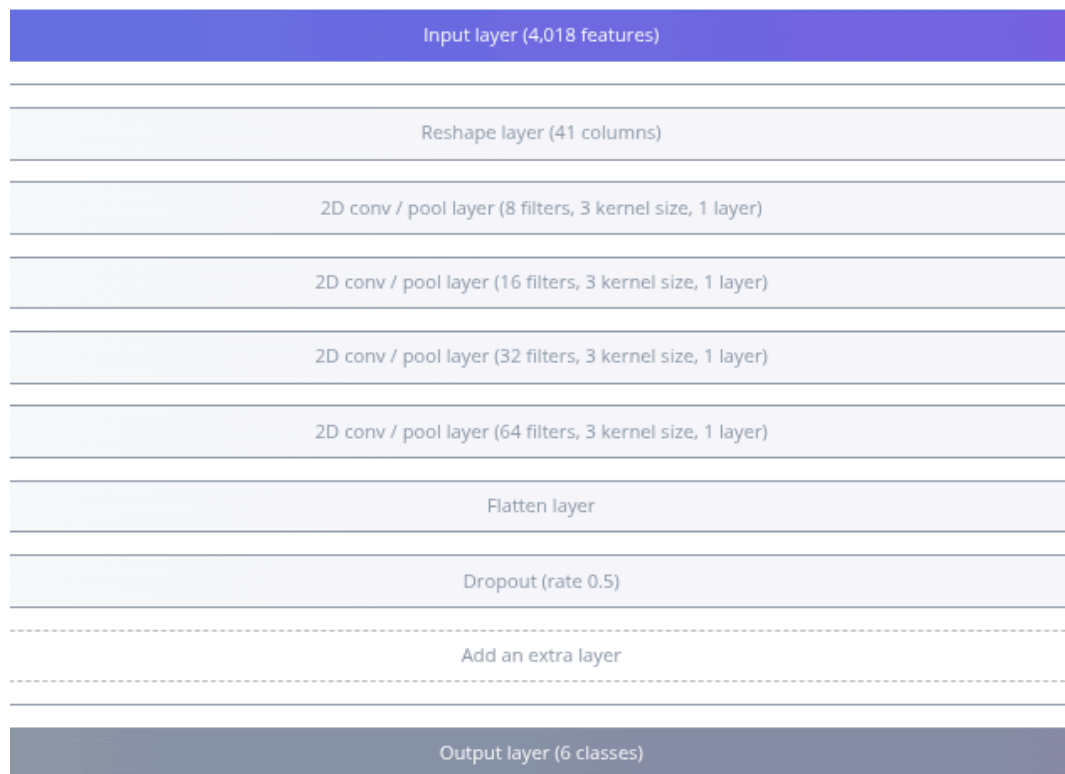
Figure 2

4.4 Model Architecture

Let the CNN be defined as a function $f_{\theta} : \mathbb{R}^{32 \times 31} \rightarrow \mathbb{R}^6$

The layers are:

- Conv2D: $W_1 \in \mathbb{R}^{3 \times 3 \times 1 \times 8}$ activation ReLU
- MaxPool2D: stride = 2
- Conv2D: $W_2 \in \mathbb{R}^{3 \times 3 \times 8 \times 16}$
- ... (repeated up to 64 filters)
- Flatten and Dense: $f_{dense}(x) = \text{softmax}(Wx + b)$



Loss function: categorical cross-entropy

$$\mathcal{L}(y, \hat{y}) = - \sum_{i=1}^6 y_i \log(\hat{y}_i)$$

Training uses **Adam optimizer** with learning rate $\eta = \mathbf{0.005}$, batch size $\mathbf{B} = 32$, and $\mathbf{E} = 80$ epochs.

5. Results

5.1 Performance Evaluation

- **Validation Accuracy:** $A_{val} = 99.5\%$
- **Loss:** $\mathcal{L}_{val} = 0.07$
- **F1 (weighted):** $F_1 = 0.99$
- **AUC:** $AUC = 1.00$

Confusion Matrix (Validation Set)

	ANGRY	DISGUST	FEAR	HAPPY	NEUTRAL	NOISE
ANGRY	100%	0%	0%	0%	0%	0%
DISGUST	0%	99.2%	0%	0.8%	0%	0%
FEAR	1.3%	0%	98.7%	0%	0%	0%
HAPPY	0%	0%	0%	100%	0%	0%
NEUTRAL	0%	0%	0%	0%	100%	0%
NOISE	0.8%	0%	0%	0%	0%	99.2%
F1 SCORE	0.99	1.00	0.99	0.99	1.00	1.00

Let TP_i , FP_i , FN_i be the true positive, false positive, and false negative for class i . Then:

$$Precision_i = \frac{TP_i}{TP_i + FP_i}, Recall_i = \frac{TP_i}{TP_i + FN_i}$$

$$F_{1i} = 2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i}$$

Average precision, recall, and F1-score for all six classes exceed **98%**.

5.1.1 Training Dynamics and Model Convergence

To evaluate model convergence and potential overfitting, accuracy and loss were tracked across 80 training epochs.

Figure 3 illustrates the progression of training and validation accuracy and loss. The model converged quickly, reaching over 95% validation accuracy within 20 epochs. Loss curves also stabilized early, with validation loss remaining consistently low across later epochs, suggesting minimal overfitting.

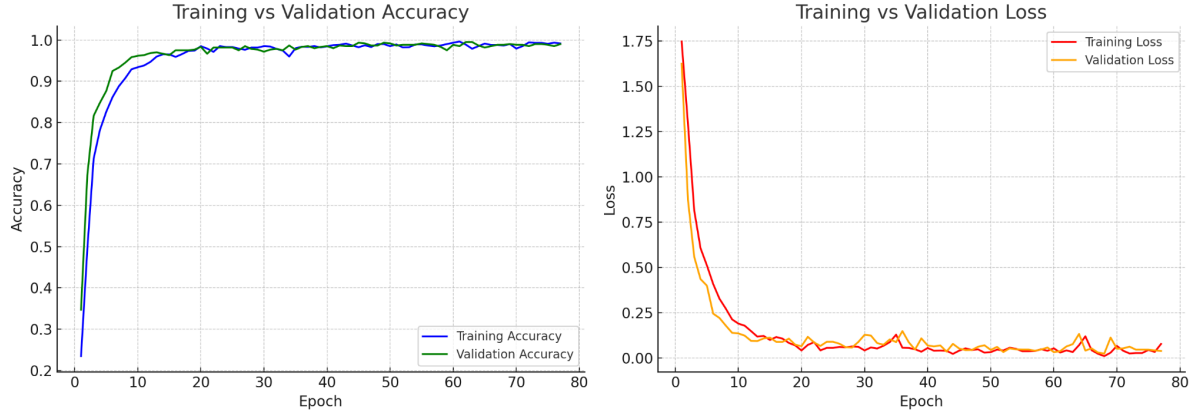


Figure 3

5.2 Resource Utilization and Performance

- **Model size (quantized):** $\approx 42.8\text{ KB}$
- **Flash size:** $\approx 1.17\text{ MB}$
- **RAM usage:** $\approx 29.6\text{ KB}$
- **DSP time:** $\mathcal{T}_{DSP} = 539\text{ ms}$
- **Classification time:** $\mathcal{T}_{infer} = 32\sim 33\text{ ms}$
- **Total inference latency:** $\mathcal{T}_{total} = \mathcal{T}_{DSP} + \mathcal{T}_{infer} = 571\sim 572\text{ ms}$

Despite the significant DSP time, the system still operates comfortably under a 1-second window, making it viable for real-time inference in low-latency applications.

5.3 Feature Space Visualization

To better understand class separability in the feature space, Edge Impulse's Data Explorer tool was used to project the high-dimensional MFE feature vectors into a 2D space using Principal Component Analysis (PCA).

Figure 4 shows the distribution of training data by class in this projected space. Clear clustering indicates that the extracted features encode emotional categories effectively, facilitating accurate classification.

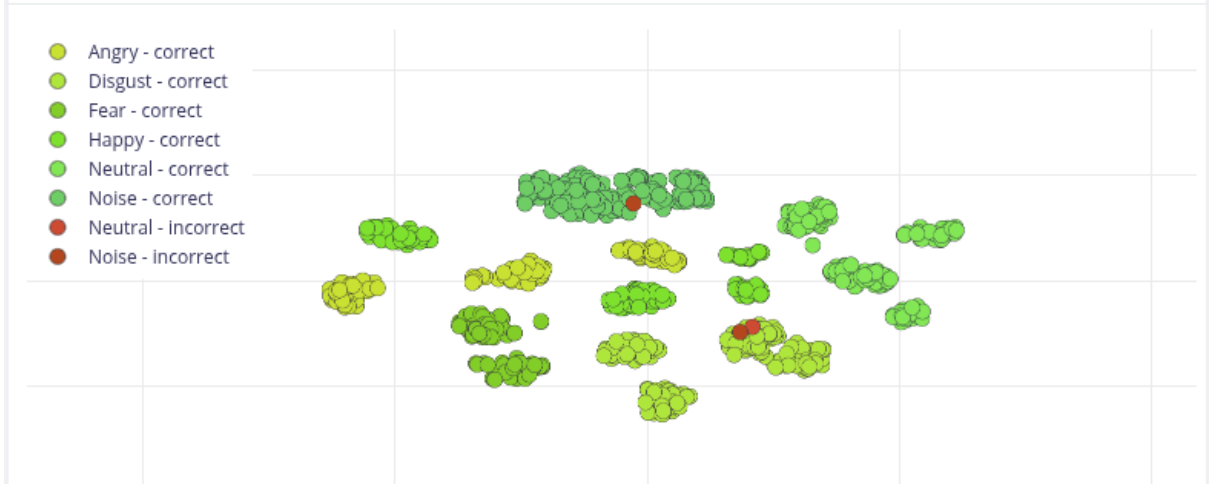


Figure 4

6. Deployment

The quantized TensorFlow Lite (int8) model was deployed to an **ESP32-PICO-D4 (dual-core, 240 MHz, 520 KB SRAM, 4 MB Flash)** Microcontroller interfaced with an **SPM1423** digital PDM microphone for real-time audio capture. Audio signals were streamed through the ESP32's I²S peripheral configured in **PDM receive mode**, processed by onboard Mel-filterbank DSP blocks, and classified by the CNN.

The I²S-PDM integration required configuring the I²S driver with:

- **Mode:** I2S_MODE_PDM
- **Sample rate:** 16,000 Hz
- **Bits per sample:** 16-bit packed
- **Buffer size:** 2048 samples per channel



This configuration enables streaming inference with minimal latency and CPU load.

Using the EON compiler, the quantized int8 model was compiled into a binary for ESP32. Inference is executed entirely offline. Audio is acquired via SPM1423 PDM I²S microphone and passed through the DSP pipeline in real-time.

$$x(t) \xrightarrow{DSP} MFE\ Matrix \xrightarrow{f_{\theta}} \hat{y}$$

7. Discussion

This experiment validates that MFE features and CNN-based classifiers can be efficiently deployed on MCUs with limited memory and computation. Despite hardware constraints, the quantized model generalizes well. Limitations include dataset size, noise robustness, and absence of speaker variation. The training and validation metrics (see Section 5.1.1) confirm that the model fits well with minimal overfitting under the given dataset. However, the drop

in real-world accuracy suggests that overfitting may occur at the feature level due to limited speaker and noise diversity.

7.1 Challenges and Limitations

Although the proposed SER model achieved a validation accuracy of 99.5% under controlled conditions, its performance in real-world scenarios showed noticeable degradation. In field testing with naturally spoken utterances recorded in less controlled environments, the model often misclassified emotions—particularly among acoustically similar categories such as *Angry* and *Disgust*. In several cases, only one or two out of five utterances per class were correctly identified, indicating a significant drop in generalization performance.

This discrepancy is likely due to a combination of the following factors:

- **Overfitting to Clean Audio:** The model was trained on clean, studio-quality samples. In contrast, real-world environments introduce unpredictable background noise, reverberation, and ambient interference that the model has not seen during training.
- **Limited Speaker Diversity:** The training dataset included a relatively narrow set of speakers. As a result, the model struggles to generalize across unfamiliar voices, especially those with varying accents, tonal qualities, or pitch ranges.
- **Environmental Noise Robustness:** Although a custom *Noise* class was included during training, real-world acoustic interference often differs significantly from the synthetic or recorded noise samples used, leading to misclassification.
- **Microphone Sensitivity and Orientation:** The onboard SPM1423 microphone, while adequate for controlled recordings, may suffer from inconsistent sensitivity depending on distance, direction, and environmental conditions, further degrading input quality.

These findings emphasize the importance of incorporating robust data augmentation techniques, collecting diverse real-world samples, and applying adaptive noise filtering to enhance the model's resilience and reliability in deployment environments.

8. Conclusion

A high-accuracy Speech Emotion Recognition (SER) model was implemented on the ESP32 using Mel-filterbank energy features and a quantized 2D Convolutional Neural Network (CNN). Achieving near real-time performance and a compact model size, this TinyML-based solution enables efficient, privacy-preserving, and fully offline emotion recognition on ultra-low-power embedded devices.

9. References

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